Architecture Validation & Quality Report

1. Executive Summary

Overall Architecture Validation Result: **SUCCESS**

2. Structural Analysis

Metric	Observed Value
Component Count	16
Relationship Count	14
Undefined References	0
Self Loops	0
Invalid Relationship Types	0

Metric Definitions and Calculation Rules
Component Count: The total number of components explicitly declared in the architectural description.
Relationship Count: The total number of relationships declared between components.
Undefined References: The number of relationships whose source or target does not match any declared component name.
Self-Loops: The number of relationships where the source and target components are identical.
Invalid Relationship Types: The number of relationships whose interaction type is not recognized by the validation rules.

Structural Evaluation Interpretation
Structural validation assesses the syntactic and referential correctness of the architecture. Any non-zero value in Undefined References, Self-Loops, or Invalid Relationship Types indicates an invalid architectural description.

3. Quality Analysis

Component	Fan-In	Fan-Out	Coupling
Hand Tracking	1	1	2
Real-Time Collaboration	1	1	2
Gaussian Blur	1	1	2
Gesture Analysis	0	1	1
Multiprocessing	1	0	1
Low-Latency	1	1	2
Place-Agnostic	1	0	1
Contour Detection	1	1	2
Background Subtraction	1	1	2
Single-Hand Interaction	1	1	2

Face Detection	0	1	1
Histogram Equalization	1	1	2
Threading	1	1	2
Face Recognition	1	1	2
Customizable Gestures	1	1	2
Gesture Recognition	1	1	2

Calculation Formulae
Fan-In(component) = Number of relationships where component is the target.
Fan-Out(component) = Number of relationships where component is the source.
Coupling(component) = Fan-In(component) + Fan-Out(component).

Quality Evaluation Interpretation
Components with high coupling are strongly dependent on other components. High coupling increases maintenance effort, change propagation, and the risk of cascading failures across the architecture.

4. Domain Analysis

Metric	Observed Value
Stateless Ratio	1.0
State Concentration	1
Communication Style	synchronous

Calculation Rules
Stateless Ratio: Inferred value representing the proportion of components without explicitly declared state.
State Concentration: Maximum Fan-In value observed across all components.
Communication Style: Determined by comparing the number of event-driven relationships to synchronous data-flow relationships.

Domain Evaluation Interpretation
High state concentration and synchronous communication patterns indicate centralized coordination and strong runtime dependencies, which negatively impact scalability, resilience, and fault tolerance.

5. Suggested Improvements

- Synchronous Communication Pattern: Components are dependent on immediate responses from other components at runtime.